



Python Dictionaries

Data with Dylan | Guided Notes | Python Fundamentals #5

Name: _____

Date: _____

- ☒ **HOW TO USE THESE NOTES:** Watch the video and fill in the blanks as Dylan explains each concept. Use the writing boxes to capture your own notes and examples. Complete the practice problem at the end!

Section 1: Introduction to Dictionaries

Topic: Storing related data in key-value pairs

What is a dictionary?

Complete the definition:

A dictionary is a Python data structure that stores data in _____ pairs.

In a real dictionary, a word is like a Python _____, while its definition is like a Python _____.

A dictionary connects a label, name, or identifier to a related piece of _____.

Key vocabulary

Complete the definitions:

A key is the _____ used to look up a piece of data.

A value is the _____ connected to a key.

A key-value pair is one key and its related _____.

Why use dictionaries?

Dictionaries are useful when information has meaningful _____ instead of only positions.

In your own words - why might names, labels, or IDs be useful as dictionary keys?

Section 2: Dictionary Syntax

Topic: Creating dictionaries with curly braces

Basic dictionary structure

Dictionaries use _____ braces: {}

A colon : separates a _____ from its value.

Commas separate multiple _____ pairs.

```
stuff = {
    "fruit": ["apples", "bananas"],
    "count": 3,
    "colors": {"red", "blue"}
}
```

Everything before the colon is a _____ .

Everything after the colon is a _____ .

Dictionary values can store lists, integers, and _____ .

Dictionary behavior

Dictionaries are _____ , meaning their values can be changed after creation.

Dictionaries are also _____ , so Python keeps pairs in insertion order.

Circle the VALID dictionary initialization:

Choice A student = {"name": "Dylan", "age": 19}	Choice B student = ["name": "Dylan", "age": 19]	Choice C student = ("name": "Dylan", "age": 19)
--	--	--

Why are the other choices not valid dictionary syntax?

Section 3: Dictionary Keys and Values

Topic: Rules for dictionary keys

Key requirements

Complete the definition:

Immutability means a value _____ be changed after it is created.

Dictionary keys must be _____ .

Keys can be strings or _____ .

Lists and sets cannot be dictionary keys because they are _____ .

Unique keys

A dictionary cannot store duplicate _____ names.

```
duplicate_stuff = {  
    "fruit": "apple",  
    "fruit": "grapes"  
}
```

Python does not raise an error, but the earlier value is _____ .

The final value connected to "fruit" will be _____ .

In your own words - why must dictionary keys be unique?

Try it yourself

What will Python display?

```
duplicate = {"fruit": "apple", "fruit": "banana"}  
print(duplicate)
```

Result: _____

Section 4: Accessing Dictionary Values

Topic: Retrieving values with keys

Using .get()

The .get() method retrieves a value when you know the _____ .

```
student_grades = {  
    "Alice": 90, "Bob": 85, "Diana": 92  
}
```

Complete the code below to retrieve Alice's grade:

```
student_grades._____("Alice")
```

What value should this return? _____

Dictionaries do not use index positions

Lists and tuples use index numbers because their items are accessed by _____ .

Dictionaries access values using _____ instead.

In your own words - what happens if you try to use an index number to grab a value instead of its key?

Complete the access examples

CODE	RESULT
student_grades.get("Bob")	_____
student_grades.get("Diana")	_____
student_grades.get("Alice")	_____

Section 5: Looping Through Dictionary Data

Topic: Working with keys and key-value pairs

Looping through keys

The `.keys()` method returns an iterable of all dictionary _____.

```
for key in student_grades:
    print(key)
```

This loop prints each student _____.

Complete the filtering logic below:

```
for key in student_grades:
    if student_grades[key] > ____:
        print(key, student_grades[key])
```

This code prints students whose grades are greater than _____.

Looping through key-value pairs

The `.items()` method returns both keys and values together as `dict_items`.

```
for key, value in student_grades.items():
    print(key, value)
```

The first item in each returned pair is the _____.

The second item in each returned pair is the `id`.

In your own words - why might `.items()` be more helpful than `.keys()` when you need both a student's name and grade?

Section 6: Updating Data and Dictionary Methods

Topic: Changing, adding, and reviewing dictionary information

Updating a value

Dictionaries are mutable, so you can update a value by using its _____ .

```
student_grades["Alice"] = 95
```

After this line runs, Alice's grade becomes _____ .

Adding a new key-value pair

```
student_grades["Evan"] = 88
```

This creates a new key called _____ with a value of _____ .

Dictionary Methods Reference

Fill in the missing method or operation names as you review each example.

CODE	METHOD / OPERATION NAME	RESULT / NOTES
student_grades.get("Alice")	_____	Returns the value connected to a known _____.
student_grades.keys()	_____	Returns an iterable of all dictionary _____.
student_grades.values()	_____	Returns an iterable of all dictionary _____.
student_grades.items()	_____	Returns key-value pairs as _____.
student_grades["Alice"] = 95	_____	Updates the value connected to "Alice".
student_grades["Evan"] = 88	_____	Adds a new _____ pair.

In your own words - when would .values() be useful?

Section 7: Putting It All Together

Topic: Dictionary reference table

Using what you have learned, complete the full data structure reference table:

DATA STRUCTURE	SYNTAX BREAKDOWN	UNIQUE TRAIT	YOUR OWN EXAMPLE
List	["apples", "bananas"]	Ordered and mutable; uses index positions.	
Set	{"apples", "bananas"}	Stores unique values without a fixed order.	
Tuple	("x", "y")	Ordered but immutable.	
Dictionary	{"name": "Dylan", "age": 19}	Stores values using unique keys.	

Section 8: Quick Recap

Test yourself - try to complete these without looking back!

1. A dictionary stores data in _____ pairs.
2. Dictionaries use _____ braces for initialization.
3. To retrieve a value, you use the dictionary's _____ instead of an index number.
4. Dictionary keys must be _____, so lists and sets cannot be keys.
5. When duplicate keys appear, Python keeps the _____ value assigned to that key.

Practice Problem

⚡ **Challenge - Pause the video and try this yourself before Dylan reveals the answer!**

```
student_grades = {"Alice": 90, "Bob": 85, "Diana": 92}
```

Goal: Correct Alice's grade so it becomes 95, then print the name and grade of every student who scored higher than 85.

Why? Why should you use "Alice" as a key instead of trying to update with an index such as 0?

Why? Why is .items() useful when printing both each student's name and grade?

Write your full statement below:

<code>student_grades["Alice"] = 95</code>

How did it go? What part tripped you up?

Personal Notes

Use this space to capture any additional observations, questions, or ideas from the video.

Key Terms Reference

Use this table to build your Python vocabulary. Write definitions in your own words.

TERM	DEFINITION IN YOUR OWN WORDS
Dictionary	
Data Structure	
Key	
Value	
Key-Value Pair	
Unique Key	
Duplicate Key	
Mutable	
Immutable	
Ordered	
Curly Braces {}	
Colon :	
Dictionary Access	
get()	

Key Terms Reference (continued)

Use this table to build your Python vocabulary. Write definitions in your own words.

TERM	DEFINITION IN YOUR OWN WORDS
keys()	
values()	
items()	
Iterable	
Tuple	
KeyError	
TypeError	
Updating a Value	
Adding a Key-Value Pair	
Assignment Operator (=)	